

Lab 1: Ethernet Cabling

Objective

This lab exercise is designed to give students hands on experience of building CAT-5 UTP Ethernet patch cables.

Resources Required

- CAT 5 Cable - bulk Category 5, 5e or 6 cable
- RJ45 Ends
- Crimper for RJ45
- Wire Cutters - to cut and strip the cable if necessary
- Wire Stripper
- Cable Tester

Introduction

Network media is the actual path over which an electrical signal travels as it moves from one component to another. There are different types of network cables such as twisted-pair cable, coaxial cable, and fiber optic cable. This section, however, describes only twisted pair and coaxial cables.

Twisted Pair Cable

Twisted-pair cable is a type of cabling that is used for telephone communications and most modern Ethernet networks. A pair of wires forms a circuit that can transmit data. The pairs are twisted to provide protection against crosstalk, the noise generated by adjacent pairs. When electrical current flows through a wire, it creates a small, circular magnetic field around the wire. When two wires in an electrical circuit are placed close together, their magnetic fields are the exact opposite of each other. Thus, the two magnetic fields cancel each other out. They also cancel out any outside magnetic fields. Twisting the wires can enhance this cancellation effect. Using cancellation together with twisting the wires, cable designers can effectively provide self-shielding for wire pairs within the network media.

Two basic types of twisted-pair cable exist: unshielded twisted pair (UTP) and shielded twisted pair (STP). The following sections discuss UTP and STP cable in more detail.

UTP Cable

UTP cable is a medium that is composed of pairs of wires (see Figure 1-1). UTP cable is used in a variety of networks. Each of the eight individual copper wires in UTP

cable is covered by an insulating material. In addition, the wires in each pair are twisted around each other.

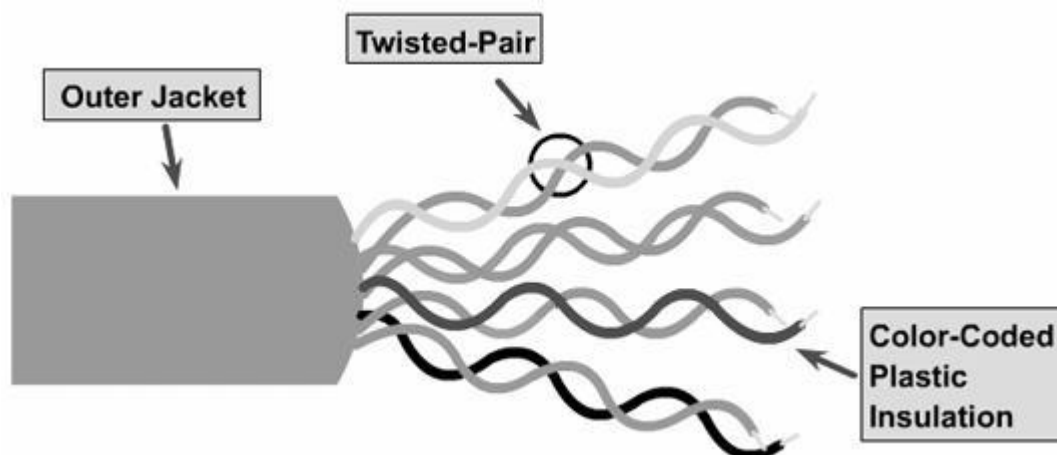


Figure 1-1 Unshielded Twisted-Pair Cable

UTP cable relies solely on the cancellation effect produced by the twisted wire pairs to limit signal degradation caused by electromagnetic interference (EMI) and radio frequency interference (RFI). To further reduce crosstalk between the pairs in UTP cable, the number of twists in the wire pairs varies. UTP cable must follow precise specifications governing how many twists or braids are permitted per meter (3.28 feet) of cable.

UTP cable often is installed using a Registered Jack 45 (RJ-45) connector (see Figure 1-2). The RJ-45 is an eight-wire connector used commonly to connect computers onto a local-area network (LAN), especially Ethernets.



Figure 1-2 RJ-45 Connectors

When used as a networking medium, UTP cable has four pairs of either 22- or 24-gauge copper wire. UTP used as a networking medium has an impedance of 100 ohms; this differentiates it from other types of twisted-pair wiring such as that used for telephone wiring, which has impedance of 600 ohms.

UTP cable offers many advantages. Because UTP has an external diameter of approximately 0.43 cm (0.17 inches), its small size can be advantageous during installation. Because it has such a small external diameter, UTP does not fill up wiring ducts as rapidly as other types of cable. This can be an extremely important factor to

consider, particularly when installing a network in an older building. UTP cable is easy to install and is less expensive than other types of networking media. In fact, UTP costs less per meter than any other type of LAN cabling. And because UTP can be used with most

of the major networking architectures, it continues to grow in popularity.

Disadvantages also are involved in using twisted-pair cabling, however. UTP cable is more prone to electrical noise and interference than other types of networking media, and the distance between signal boosts is shorter for UTP than it is for coaxial and fiber-optic cables.

Although UTP was once considered to be slower at transmitting data than other types of cable, this is no longer true. In fact, UTP is considered the fastest copper-based medium today.

The following summarizes the features of UTP cable:

- Speed and throughput—10 to 1000 Mbps
- Average cost per node—Least expensive
- Media and connector size—Small
- Maximum cable length—100 m (short)

Commonly used types of UTP cabling are as follows:

Category 1—Used for telephone communications. Not suitable for transmitting data.

Category 2—Capable of transmitting data at speeds up to 4 megabits per second (Mbps).

Category 3—Used in 10BASE-T networks. Can transmit data at speeds up to 10 Mbps.

Category 4—Used in Token Ring networks. Can transmit data at speeds up to 16 Mbps.

Category 5—Can transmit data at speeds up to 100 Mbps.

Category 5e—Used in networks running at speeds up to 1000 Mbps (1 gigabit per second [Gbps]).

Category 6—Typically, Category 6 cable consists of four pairs of 24 American Wire Gauge (AWG) copper wires. Category 6 cable is currently the fastest standard for UTP.

Shielded Twisted-Pair Cable

Shielded twisted-pair (STP) cable combines the techniques of shielding, cancellation, and wire twisting. Each pair of wires is wrapped in a metallic foil (see Figure 1-3). The four pairs of wires then are wrapped in an overall metallic braid or foil, usually 150-ohm cable. As specified for use in Ethernet network installations, STP reduces electrical noise both within the cable (pair-to-pair coupling, or crosstalk) and from outside the cable (EMI and RFI). STP usually is installed with STP data connector, which is created especially for the STP cable. However, STP cabling also can use the same RJ connectors that UTP uses.

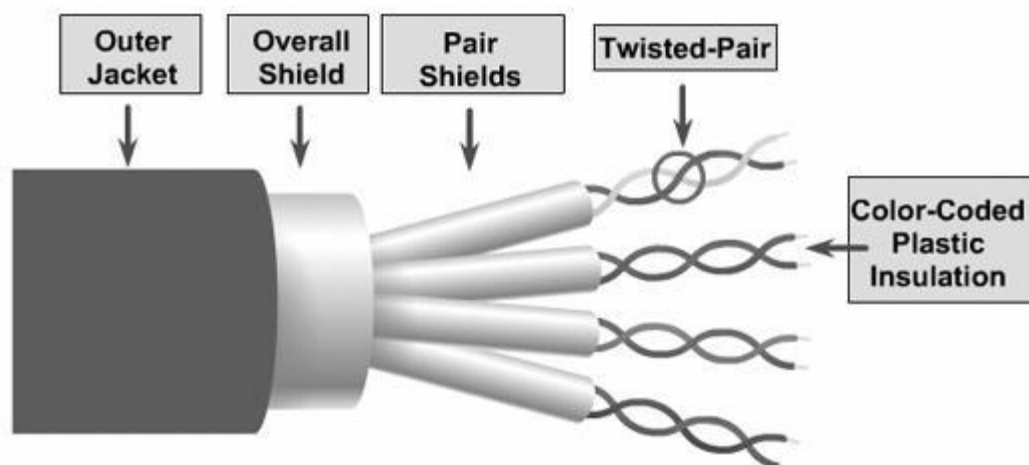


Figure 1-3 Shielded Twisted-Pair Cable

Although STP prevents interference better than UTP, it is more expensive and difficult to install. In addition, the metallic shielding must be grounded at both ends. If it is improperly grounded, the shield acts like an antenna and picks up unwanted signals. Because of its cost and difficulty with termination, STP is rarely used in Ethernet networks. STP is primarily used in Europe.

The following summarizes the features of STP cable:

- Speed and throughput—10 to 100 Mbps
- Average cost per node—Moderately expensive
- Media and connector size—Medium to large
- Maximum cable length—100 m (short)

When comparing UTP and STP, keep the following points in mind:

- The speed of both types of cable is usually satisfactory for local-area distances.
- These are the least-expensive media for data communication. UTP is less expensive than STP.
- Because most buildings are already wired with UTP, many transmission standards are adapted to use it, to avoid costly rewiring with an alternative cable type.

Coaxial Cable

Coaxial cable consists of a hollow outer cylindrical conductor that surrounds a single inner wire made of two conducting elements. One of these elements, located in the center of the cable, is a copper conductor. Surrounding the copper conductor is a layer of flexible insulation. Over this insulating material is a woven copper braid or metallic foil that acts both as the second wire in the circuit and as a shield for the

inner conductor. This second layer, or shield, can help reduce the amount of outside interference. Covering this shield is the cable jacket.

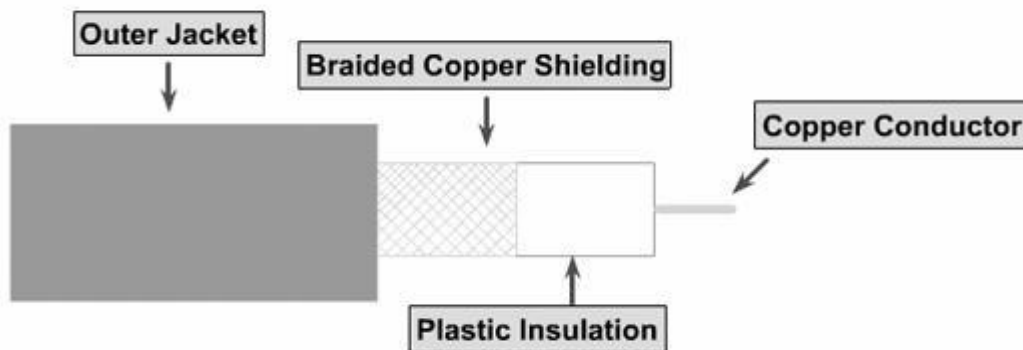


Figure 1-4 Coaxial Cable

Coaxial cable supports 10 to 100 Mbps and is relatively inexpensive, although it is more costly than UTP on a per-unit length. However, coaxial cable can be cheaper for a physical bus topology because less cable will be needed. Coaxial cable can be cabled over longer distances than twisted-pair cable. For example, Ethernet can run approximately 100 meters (328 feet) using twisted-pair cabling. Using coaxial cable increases this distance to 500m (1640.4 feet).

For LANs, coaxial cable offers several advantages. It can be run with fewer boosts from repeaters for longer distances between network nodes than either STP or UTP cable. Repeaters regenerate the signals in a network so that they can cover greater distances. Coaxial cable is less expensive than fiber-optic cable, and the technology is well known; it has been used for many years for all types of data communication.

When working with cable, you need to consider its size. As the thickness, or diameter, of the cable increases, so does the difficulty in working with it. Many times cable must be pulled through existing conduits and troughs that are limited in size. Coaxial cable comes in a variety of sizes. The largest diameter (1 centimeter [cm]) was specified for use as Ethernet backbone cable because historically it had greater transmission length and noise-rejection characteristics. This type of coaxial cable is frequently referred to as Thicknet. As its nickname suggests, Thicknet cable can be too rigid to install easily in some situations because of its thickness. The general rule is that the more difficult the network medium is to install, the more expensive it is to install. Coaxial cable is more expensive to install than twisted-pair cable. Thicknet cable is almost never used except for special-purpose installations.

A connection device known as a vampire tap was used to connect network devices to Thicknet. The vampire tap then was connected to the computers via a more flexible cable called the attachment unit interface (AUI). Although this 15-pin cable was still thick and tricky to terminate, it was much easier to work with than Thicknet.

In the past, coaxial cable with an outside diameter of only 0.35 cm

(sometimes referred to as Thinnet) was used in Ethernet networks. Thinnet was especially useful for cable installations that required the cable to make many twists and turns. Because it was easier to install, it was also cheaper to install. Thus, it was sometimes referred to as Cheapernet. However, because the outer copper or metallic braid in coaxial cable comprises half the electrical circuit, special care had to be taken to ensure that it was properly grounded. Grounding was done by ensuring that a solid electrical connection existed at both ends of the cable. Frequently, however, installers failed to properly ground the cable. As a result, poor shield connection was one of the biggest sources of connection problems in the installation of coaxial cable. Connection problems resulted in electrical noise, which interfered with signal transmittal on the networking medium. For this reason, despite its small diameter, Thinnet no longer is commonly used in Ethernet networks.

The most common connectors used with Thinnet are BNC, short for British Naval Connector or Bayonet Neill Concelman, connectors (see Figure 1-5). The basic BNC connector is a male type mounted at each end of a cable. This connector has a center pin connected to the center cable conductor and a metal tube connected to the outer cable shield. A rotating ring outside the tube locks the cable to any female connector. BNC Tconnectors are female devices for connecting two cables to a network interface card (NIC). A BNC barrel connector facilitates connecting two cables together.



Figure 1-5 Thinnet and BNC Connector

The following summarizes the features of coaxial cables:

- Speed and throughput—10 to 100 Mbps
- Average cost per node—Inexpensive
- Media and connector size—Medium
- Maximum cable length—500 m (medium)

Internal Structure of the UTP Cable

Here is what the internals of the cable look like:



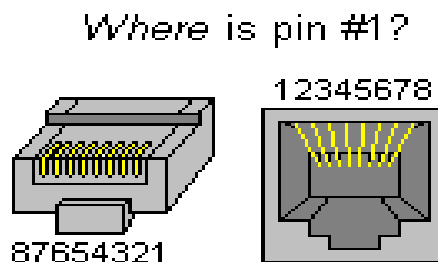
Internal Cable Structure and Color Coding

Inside the cable, there are 8 color coded wires. These wires are twisted into 4 pairs of wires; each pair has a common color theme. One wire in the pair being a solid or primarily solid colored wire and the other being a primarily white wire with a colored stripe (Sometimes cable doesn't have any color on the striped cable, the only way to tell is to check which other wire it is twisted around). Examples of the naming schemes used are: Orange (alternatively Orange/White) for the solid colored wire and White/Orange for the striped cable. The twists are extremely important. They are there to counteract noise and interference. It is important to wire according to a standard to get proper performance from the cable. The TIA/EIA-568-A specifies two wiring standards for a 8-position modular connector such as RJ45. The two wiring standards, T568A and T568B vary only in the arrangement of the colored pairs.

RJ45 Ends

The RJ45 end is an 8-position modular connector that looks like a large phone plug. There are a couple variations available. The primary variation you need to pay attention to is whether the connector is intended for braided or solid wire. For braided/stranded wires, the connector has contacts that actually pierce the wire. For solid wires, the connector has fingers which pierce the insulation and make contact with the wire by grasping it from both sides.

Here is a diagram and pinout:

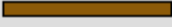


RJ45 Jack and Plug Pinout

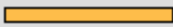
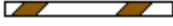
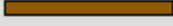
Ethernet Cable Pinouts:

There are two basic cables. A straight through cable, which is used to connect to a hub or switch, and a cross over cable used to operate in a peer-to-peer fashion without a hub/switch. Some interfaces can cross and un-cross a cable automatically as needed, really quite nice.

Standard, Straight-Through Wiring (both ends are the same)

RJ45 Pin #	Wire Color (T568A)	Wire Diagram (T568A)	10Base-T Signal 100Base-TX Signal	1000Base-T Signal
1	White/Green		Transmit+	BI_DA+
2	Green		Transmit-	BI_DA-
3	White/Orange		Receive+	BI_DB+
4	Blue		Unused	BI_DC+
5	White/Blue		Unused	BI_DC-
6	Orange		Receive-	BI_DB-
7	White/Brown		Unused	BI_DD+
8	Brown		Unused	BI_DD-

Straight-Through Cable Pinout for T568A

RJ45 Pin #	Wire Color (T568B)	Wire Diagram (T568B)	10Base-T Signal 100Base-TX Signal	1000Base-T Signal
1	White/Orange		Transmit+	BI_DA+
2	Orange		Transmit-	BI_DA-
3	White/Green		Receive+	BI_DB+
4	Blue		Unused	BI_DC+
5	White/Blue		Unused	BI_DC-
6	Green		Receive-	BI_DB-
7	White/Brown		Unused	BI_DD+
8	Brown		Unused	BI_DD-

Straight-Through Cable Pinout for T568B

Procedure to make Ethernet Cables:

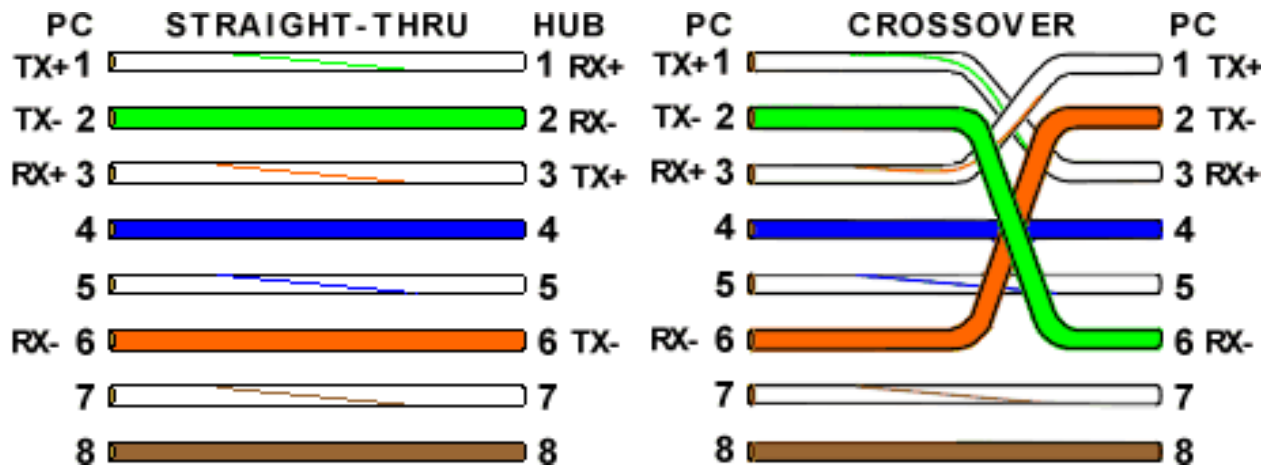
1. Strip off about 2 inches of the cable sheath.
2. Untwist the pairs - don't untwist them beyond what you have exposed, the more untwisted cable you have the worse the problems you can run into.
3. Align the colored wires according to the diagrams above.
4. Trim all the wires to the same length, about 1/2" to 3/4" left exposed from the sheath.
5. Insert the wires into the RJ45 end - make sure each wire is fully inserted to the front of the RJ45 end and in the correct order. The sheath of the cable should extend into the RJ45 end by about 1/2" and will be held in place by the crimp.
6. Crimp the RJ45 end with the crimper tool



Crimper tool

7. Verify the wires ended up the right order and that the wires extend to the front of the RJ45 end and make good contact with the metal contacts in the RJ45 end.
8. Cut the cable to length - make sure it is more than long enough for your needs. Remember, an end to end connection should not extend more than 100m (~328ft). Try to keep cables short, the longer the cable becomes the more it may affect performance, usually noticeable as a gradual decrease in speed and increase in latency.
9. Repeat the above steps for the second RJ45 end.
10. If a cable tester is available, use it to verify the proper connectivity of the cable.

That should be it, if your cable doesn't turn out, look closely at each end and see if you can find the problem. Usually a wire ended up in the wrong place or more commonly, one of the wires didn't extend to the front of the RJ45 connector and is making no, or poor contact. If you see a mistake or problem, cut the end off and start again.



Straight thru and Crossover connection diagram

Lab Tasks**Post Lab Task 1:** Answer the following:

- i. What is the advantage of having twists in a twisted pair cable?

- ii. Identify the differences between twisted pair and coaxial cables.

- iii. Which type twisted pair cable will you use to connect? Router to Hub, Router to Router, PC to HUB, PC to PC

- iv. Can you connect a hub to another hub or a switch to another switch to another switch using a straight-through cable? Explain your answer.

- v. Summarize in tabular form the features of UTP, SPT and Coaxial cables in terms of Speed, Average code per node, connector size and Maximum cable length?

- vi. What is difference between thicknet and thinnet? Write connectors name of thicknet and thinnet?
